

D-ary Codes

Problem ID: problem-36-23810308

You have unearthed a repository of classic texts from the planet Earth. You are certain these texts have great archeological (and financial) value. Unfortunately, as all forms of digital storage were outlawed after the AI apocalypse, your storage capacity is very limited.

Moreover, while the source text was encoded using binary 0s and 1s, your collection of black-market storage devices use *many-valued logic*. For example, your ternary computer drive uses ternary digits, a.k.a ‘trits’, where each value can be 0, 1 or 2.

Can you find the optimal variable-length code for compressing a given text using d distinct symbols?

Input

The first line contains two integers n and d , where n is the number of lines in the text, and d is the number of symbols used by your storage drive. Each of the next n lines contains between 2 and 80 Unicode characters, where the length includes the line ending `\n`.

Problem bounds are $1 \leq n \leq 10^4$ and $3 \leq d \leq 10$. The maximum number of characters is 80×10^4 . Additionally, the number of unique characters k in any given text, including `\n`, is bounded by $2 \leq k \leq 100$.

Important: New lines `\n` are part of the text and should be included in the encoding. On Windows, line endings are represented by two characters, `\r\n`. To avoid any operating system-specific issues, only count line feed characters `\n`. Any carriage return characters `\r` should be ignored.

Output

Output the length of the encoded text using an optimal variable-length code for an alphabet of size d .

Sample Input 1

```
1 3
Fair is foul, and foul is fair.
```

Sample Output 1

```
73
```

Sample Input 2

```
2 3
Remember, that I am thy creature: I ought to be thy Adam; but I am rather the
fallen angel, whom thou drivest from joy for no misdeed.
```

Sample Output 2

```
374
```